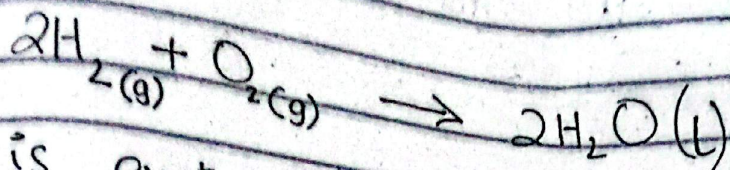
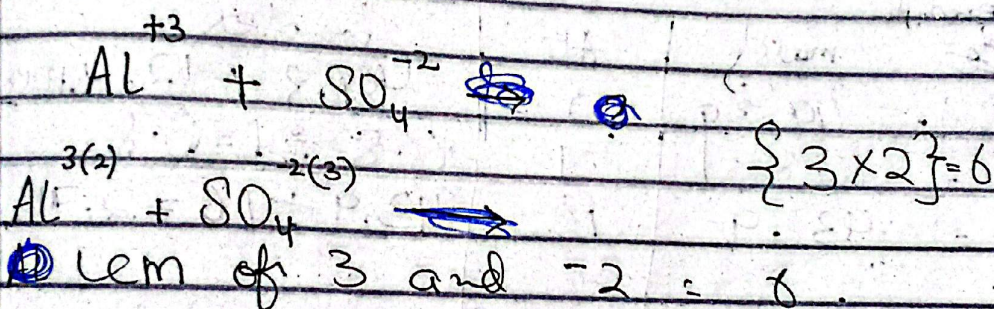


Q:2(i)



It is exothermic reaction,
reason:- Breaking bonds in the reactants
O-H bonds, while forming the new
more energy in water release energy.
bonds form is released when the new
the original bonds than is absorbed to break
the original bonds.

Q:2(ii)

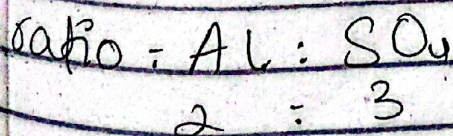


for aluminium:-

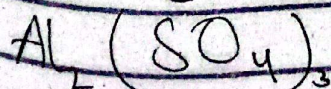
$$\frac{6}{3} = 2$$

for sulphate:-

$$\frac{6}{2} = 3$$



criss cross :-



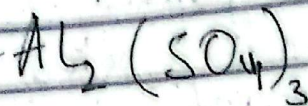
Check ~~the~~ charges :-

$$= 2(+3) + 2(-2)$$

$$= 6 - 4$$

$$= 2$$

formula for aluminium sulfate



Q:3
 (a) Empirical formula -
 consider, we have 100 g of compound

Element	mass	At	moles
C	49.3g	12	$49.3 \div 12 = 4.108$
H	6.8g	1	$6.8 \div 1 = 6.8$
O	43.9	16	$43.9 \div 16 = 2.744$

Now we will divide by the smallest values of moles :-

$$\text{C} :- \frac{4.108}{2.744} = 1.497 \approx 1.5$$

$$\text{H} :- \frac{6.8}{2.744} = 2.478 \approx 2.5$$

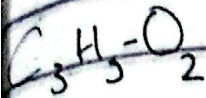
$$\text{O} :- \frac{2.744}{2.744} = 1 \approx 1$$

Ratio:-

$$1.5 : 2.5 : 1$$

multiply by 2 :-

$$3 : 5 : 2$$



② molecular formula if
 $C_3H_5O_2$:-

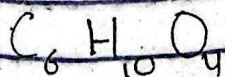
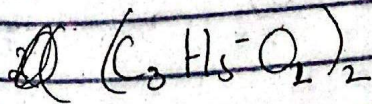
$$3(12) + 5(1) + 2(16)$$

$$36 + 5 + 32 = 73$$

Given, Molecular {relative} = 146

$$\frac{146}{73} = 2$$

multiply the empirical formula
by 2 :-



moles in 43.8 g

$$n = \frac{\text{mass}}{M_r}$$

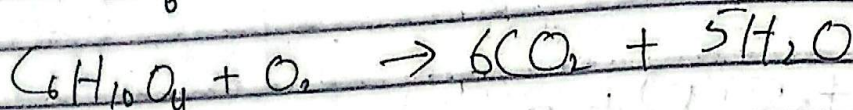
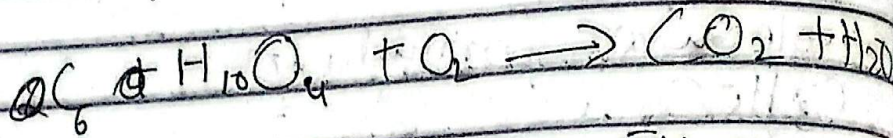
$$= \frac{43.8}{146} \Rightarrow 0.300 \text{ mol}$$

(d) number of molecules

$$\text{molecules} = n \times N_A$$

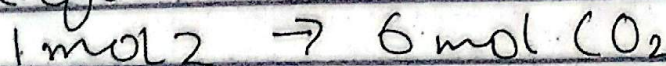
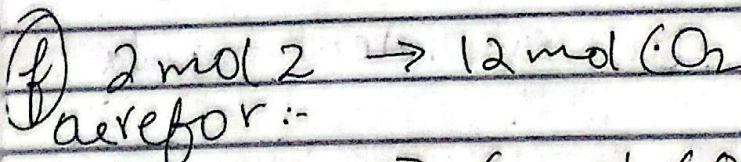
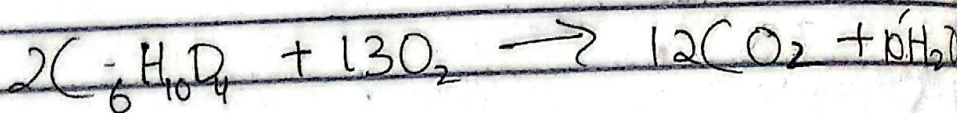
$$0.300 \times 6.022 \times 10^{23} \\ = 1.807 \times 10^{23}$$

(e) Complete combustion :-



$$6 \times 2 + 5 \times 1 = 17$$

There are 4 O atoms on the left we need 13 more.



So,

$$0.3 \times 6 = 1.8 \text{ mol } CO_2$$

molar mass of CO_2 :-

$$12 + (2 \times 16)$$

$$44 \text{ g/mol}$$

$$\text{mass} = 1.8 \times 44 \\ = 79.2$$

MCQs

30. B.

1. A
2. B
3. C
4. D
5. B
6. A
7. D
8. C
9. B
10. B
11. D
12. C
13. A
14. B
15. C
16. D
17. A
18. B
19. A
20. D
21. B
22. A
23. P
24. C
25. B
26. A
27. B
28. D
29. A